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SciScore Report

Below you will find your SciScore report containing three tables. Your score is calculated based on adherence to scientific rigor criteria (Table 1) and identification of key biological resources (Table 2). Table 3 contains statistical tests and oligonucleotides but is not scored. If SciScore makes any mistakes, please contact us to help us learn and improve.

Table 1: Rigor Adherence Table

Ethics
Field Sample Permit: Human sample collection for research was conducted in accordance with the recognized ethical guideline of Declaration of Helsinki and approved by the institutional Ethics Committee of the University Hospital of Liege (Liege, Belgium; file #B707201111974).
IRB: Human sample collection for research was conducted in accordance with the recognized ethical guideline of Declaration of Helsinki and approved by the institutional Ethics Committee of the University Hospital of Liege (Liege, Belgium; file #B707201111974).
Inclusion and Exclusion Criteria
not detected.
Attrition
not detected.
Sex as a biological variable
not detected.
Subject Demographics
Age: 2.3 Mouse models In vivo tumor growth and regrowth (relapse) after TKIs (sunitinib or sorafenib) treatment cessation [19] and the effect of inhibitors of SCD1 (SCD1 Inh) or FABP4 (FABP4 Inh) Inh were evaluated on MDA-M231 xenografts grown in immunodeficient RAG1–/– mice (6–8 weeks old) (The Jackson Laboratory) or syngeneic LLC tumor grown in C57bl/6 (6–8 weeks old) mice (Charles River).
Randomization
not detected.
Blinding
not detected.
Power Analysis
not detected.

Replication
Number: Representative results of one experiment out of three independent experiments are shown.
Cell Line Authentication
Contamination: Cell line authentication for interspecies contamination, was performed by Leibniz-Institute DSMZ (GmbH, Braunschweig, Germany).

Table 2: Key Resources Table

Your Sentences	REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies			
Antigenic bands were detected by exposing the membranes to human primary antibodies targeting the following proteins: SCD1 [M38], SCD1 [ab52356, Abcam], TxNIP [D5F3E], Cell Signaling (Beverly, MA, USA), FABP4 [ab13979], GPX4 [ab231174] Abcam (Cambridge, England), SOD1 [8B10], Catalase [PA5-23246] Thermofisher (Rockford, IL, USA), GPX1 [A0873] and serotransferrin [A1448] (Abclonal, Woburn, MA, USA).	SCD1		
	TxNIP [D5F3E], Cell Signaling		
	FABP4		
	SOD1		
	GPX1		
After washes, membranes were incubated with a secondary horseradish peroxydase (HRP)-conjugated goat anti-rabbit antibody (1:2000, DakoCytomation) or sheep anti-mouse antibody (1:1000, DakoCytomation).	anti-rabbit		
	anti-mouse		
For loading control, membranes were stripped and re-incubated with Hsc70 [B-6, sc-7298] antibody (Santa Cruz, CA, USA), or GAPDH [MAB37A] and actin [A2066] antibody (Sigma Aldrich, Saint-Louis, USA).	Hsc70		
	GAPDH		
	actin [A2066		
2.12 Immunohistochemistry Primary antibodies and dilutions used are: Rabbit polyclonal anti-Lyve-1 (1/1000) (Bioconnect, Huissen, Netherlands); rabbit polyclonal anti-Fatty acid synthase (FASN) (1/100) (Cell signaling, #3180); rabbit polyclonal anti-perilipin A (1/500) (ab352, Abcam); rabbit polyclonal anti-carbonic anhydrase 9 (CA-IX) (1/1000) (ab15086, Abcam); mouse monoclonal anti-CD31 (1/40) (ab10559 Abcam); mouse polyclonal anti-SCD1 (1/100) (ab19862, Abcam); FABP4 (1/1200) (ab13979); PLIN2 (1/400) (ab52356, Abcam), Pimonidazole (1/50), GPX4 (1/500) (ab231174, Abcam) and 4-hydroxynonenal (HNE) at 1/500 (ab46545).	anti-Lyve-1		
	anti-CD31		
	GPX4		
After washes in phosphate buffered saline (PBS), the appropriate secondary antibodies conjugated to peroxidase or biotin for 30 min were	anti-rat/biotin		
	anti-rabbit/biotin		

added: rabbit anti-rat/biotin (diluted 1/300, DAKO), goat anti-rabbit/ biotin (diluted 1/400, DAKO), rabbit anti-guinea pig/HRP (diluted 1/50, DAKO), rabbit anti-rat/HRP (diluted 1/400, DAKO) and goat anti-rabbit ENVISION/HRP, goat anti mouse ENVISION/HRP.	anti-guinea pig/ HRP		
	anti-rat/HRP		
	anti mouse ENVISION/HRP		
Cells were incubated with anti-HNE antibody (ab46545, Abcam) at 1/500 in 1% BSA/PBS for 1 h at RT.	anti-HNE		
Experimental Models: Cell Lines			
2 Materials and methods 2.1 Cell lines Human breast cancer (MDA-MB-231) and mouse Lewis Lung Carcinoma (LLC) were purchased from American Type Culture Collection (ATCC) (Manassas, USA).	MDA-MB-231		Suggestion: RRID:CVCL_0062(link)
Human Umbilical Vein Endothelial Cells (HUVEC) were purchased from Lonza.	HUVEC		Suggestion: RRID:CVCL_2959(link)
Cancer cells were grown in Dulbecco's modified Eagles's medium (DMEM) and HUVECs were cultured in EBM-2 basal medium (Lonza, CC-3156) and EGM-2 SingleQuots supplements (Lonza, CC-4176).	HUVECs		Suggestion: RRID:CVCL_2959(link)
2.3 Mouse models In vivo tumor growth and regrowth (relapse) after TKIs (sunitinib or sorafenib) treatment cessation [19] and the effect of inhibitors of SCD1 (SCD1 Inh) or FABP4 (FABP4 Inh) Inh were evaluated on MDA-M231 xenografts grown in immunodeficient RAG1–/– mice (6–8 weeks old) (The Jackson Laboratory) or syngeneic LLC tumor grown in C57bl/6 (6–8 weeks old) mice (Charles River).	MDA-M231		
For cisplatin, BALB/c mice bearing 4T1 tumors (50–100 mm^3) were administered by i.p. injection of vehicle (0.7% DMSO in PBS) or cisplatin (7 mg/kg/week) for 3 weeks.	4T1		
Mice bearing tumors were treated with TKIs (sunitinib: 40 mg/kg/day or sorafenib: 20 mg/kg/day) for 30 days for MDA-MB231 xenografts and 15 days for LLC tumors followed by the drug washout period (re-oxygenation phase) that was 3 and 2 weeks for the MDA-MB23 and LLC tumors, respectively.	MDA-MB231		Suggestion: RRID:CVCL_0062(link)
	MDA-MB23		

2.5 shRNA transduction MDA-MB-231 cells were transduced with shRNA lentiviral vectors containing Non-targeted shRNA (NT-ShRNA) (anti-eGFP) or shRNA directed against SCD1 (TRCN#00000337876 = shRNA n°1, TRCN#00003378757875 = shRNA n °2) (Sigma-Aldrich, St-Louis, USA).	MDA-MB-231 cells		
MDA-MB231 and LLC cells (25 × 103) were incubated in DMEM supplemented with 10% FBS OVN.	LLC		
2.13 Immunofluorescent detection of HNE Detection of 4-hydroxynonenal (HNE) in cancer cells in vitro was performed on coverslips of sh-SCD1 or shNT MDA-MB231 cells incubated or not with H2O2 (100 µM) for 30 min.	MDA-MB231 cells		
Experimental Models: Organisms/Strains			
2.3 Mouse models In vivo tumor growth and regrowth (relapse) after TKIs (sunitinib or sorafenib) treatment cessation [19] and the effect of inhibitors of SCD1 (SCD1 Inh) or FABP4 (FABP4 Inh) Inh were evaluated on MDA-M231 xenografts grown in immunodeficient RAG1–/– mice (6–8 weeks old) (The Jackson Laboratory) or syngeneic LLC tumor grown in C57bl/6 (6–8 weeks old) mice (Charles River).	RAG1–/–		
For cisplatin, BALB/c mice bearing 4T1 tumors (50–100 mm^3) were administered by i.p. injection of vehicle (0.7% DMSO in PBS) or cisplatin (7 mg/kg/week) for 3 weeks.	BALB/c		Suggestion: RRID:MGI:2683685(link)
Software and Algorithms			
Image processing and measurements were performed using the toolbox of image analysis of the MATLAB 9.4 (R2018a) software (Mathworks, Inc.). 2.21 Statistical analysis Considering in vitro experimentation, differences between experimental groups were assessed using a Mann-Whitney test using Prism 7.0 software (GraphPad, San Diego, CA).	MATLAB		Suggestion: MATLAB, RRID:SCR_001622(link)
	Prism		Suggestion: Meta-Prism (RRID:SCR_021836)(link)
	GraphPad		Suggestion: GraphPad Prism, RRID:SCR_002798(link)
In vivo, to compare differences between mice treatments we used a multiple comparison ANOVA two ways with Holm-Sidak parameter using SigmaPlot software (SYSTAT, San Jose, CA).	SigmaPlot		

Other Entities Detected

Your Sentences	Recognized Entity
Statistical Tests	
Image processing and measurements were performed using the toolbox of image analysis of the MATLAB 9.4 (R2018a) software (Mathworks, Inc.). 2.21 Statistical analysis Considering in vitro experimentation, differences between experimental groups were assessed using a Mann-Whitney test using Prism 7.0 software (GraphPad, San Diego, CA).	Mann-Whitney test
A non-parametric Mann-Whitney significance test was used to compare the median values measured for the groups.	Mann-Whitney significance test
In vivo, to compare differences between mice treatments we used a multiple comparison ANOVA two ways with Holm-Sidak parameter using SigmaPlot software (SYSTAT, San Jose, CA).	ANOVA

SciScore is an automated tool that is designed to assist expert reviewers by finding and presenting formulaic information scattered throughout a paper in a standard, easy to digest format. ***SciScore is not a substitute for expert review.*** SciScore also checks for the presence and correctness of several unique identifiers, including RRIDs (research resource identifiers) in the manuscript, detects sentences that appear to be missing RRIDs, and can even suggest RRIDs under certain circumstances. **All RRID suggestions should be verified;** only the author can know whether the suggestions are correct.

For a full description of scored criteria and tips for improving your score, please see https://www.scicrunch.com/sciscorereport-faq